NITTE (Deemed to be University) | NMAM INSTITUTE OF TECHNOLOGY

(An off-Campus Institution of NITTE (DEEMED TO BE UNIVERSITY), MANGALORE)

Department of MCA

07-04-2026

ADVANCED WEB TECHNOLOGIES LAB ASSIGNMENT 6

1. Write a JavaScript program that tracks user interaction with a textbox using: focus, blur, change, key up. Display the event logs dynamically on the webpage.

```
<!DOCTYPE html>
<html>

<body>

  <input id="box" placeholder="Type something...">
  <ul id="log"></ul>

  <script>
    const box = document.getElementById("box"),
      log = document.getElementById("log");

    function add(e) {
      log.innerHTML = `<li>${e.type} : ${box.value}</li>` + log.innerHTML;
    }

    ["focus", "blur", "change", "keyup"].forEach(ev =>
      box.addEventListener(ev, add)
    );
  </script>

</body>

</html>
```

Output:

hello

- keyup : hello
- keyup : hell
- keyup : hel
- keyup : he
- keyup : h
- focus :

2. Write a JavaScript program that accepts a paragraph in a text area and allows the user to:
i. Search a word or pattern
ii. Count occurrences
iii. Replace the word with another word
iv. Use regular expressions, string methods, and DOM manipulation.

```
<!DOCTYPE html>
<html>

<body>

<textarea id="txt" rows="6" cols="50" placeholder="Enter paragraph..."></textarea><br><br>

  <input id="find" placeholder="Search / Pattern">
  <input id="rep" placeholder="Replace with">
  <label><input type="checkbox" id="useRegex"> Use Regex</label>
  <button onclick="run()">Run</button>

  <p id="out"></p>

  <script>
    function run() {
      let t = document.getElementById("txt").value,
        f = document.getElementById("find").value,
        r = document.getElementById("rep").value,
        useRegex = document.getElementById("useRegex").checked;
```

```
        if (!f) {
          document.getElementById("out").innerHTML = "Enter search text";
          return;
        }

        let re;
        try {
          // if not regex mode → escape input
          let pattern = useRegex ? f : f.replace(/[.*+?^${}()|[\]\\]/g, "\\$&");
          re = new RegExp(pattern, "gi");
        } catch (e) {
          document.getElementById("out").innerHTML = "Invalid pattern";
          return;
        }

        let matches = t.match(re),
          count = matches ? matches.length : 0,
          replaced = t.replace(re, r);

        document.getElementById("out").innerHTML =
          "Count: " + count + "<br><br>" +
          "Replaced:<br>" + replaced;
      }
    </script>

</body>

</html>
```

Output:

Enter paragraph...

Search / Pattern | Replace with | ☐ Use Regex | Run

Apple is good. I like apple. APPLE is healthy.

apple | mango | ☐ Use Regex | Run

Count: 3

Replaced:
mango is good. I like mango. mango is healthy.

ant apple bat ball axe anchor

a\w+ | s | ☑ Use Regex | Run

Count: 6

Replaced:
s s bs bs s s

3. Using an implicit array, display the number of checkboxes related to grocery items to be purchased have been checked.

```
<!DOCTYPE html>
<html>

<body>

  <form>
    <label><input type="checkbox" name="g"> Rice</label>
    <label><input type="checkbox" name="g"> Milk</label>
    <label><input type="checkbox" name="g"> Eggs</label>
```

```
    <label><input type="checkbox" name="g"> Bread</label><br><br>

    <button type="button" onclick="count(this)">Count Checked</button>
  </form>

  <p id="out"></p>

  <script>
    function count(btn) {
      let c = 0, items = btn.form.elements; // implicit array
      for (let i = 0; i < items.length; i++)
        if (items[i].type == "checkbox" && items[i].checked) c++;

      out.innerHTML = "Checked items: " + c;
    }
  </script>

</body>

</html>
```

Output:

☐ Rice ☐ Milk ☐ Eggs ☐ Bread

Count Checked

☐ Rice ☑ Milk ☑ Eggs ☐ Bread

Count Checked

Checked items: 2

☑ Rice ☑ Milk ☑ Eggs ☑ Bread

Count Checked

Checked items: 4

4. Create an HTML document that includes five radio buttons, labelled red, blue, green, yellow, and orange. The event handlers for these buttons must produce messages stating the chosen favourite color. The event handler must be implemented as a function whose name must be assigned to the onclick attribute of the radio button elements. The chosen color must be sent to the event handler as a parameter.

```
<!DOCTYPE html>
<html>

<body>

  <p>Select your favourite color:</p>

  <label><input type="radio" name="c" onclick="show('Red')"> Red</label>
  <label><input type="radio" name="c" onclick="show('Blue')"> Blue</label>
  <label><input type="radio" name="c" onclick="show('Green')"> Green</label>
  <label><input type="radio" name="c" onclick="show('Yellow')"> Yellow</label>
  <label><input type="radio" name="c" onclick="show('Orange')"> Orange</label>

  <p id="out"></p>

  <script>
    function show(color) {
      out.innerHTML = "Your favourite color is " + color;
    }
  </script>

</body>

</html>
```

<u>Output:</u>

Select your favourite color:

○ Red ○ Blue ○ Green ○ Yellow ○ Orange

Select your favourite color:

○ Red ○ Blue ○ Green ○ Yellow ◉ Orange

Your favourite color is Orange

Select your favourite color:

○ Red ◉ Blue ○ Green ○ Yellow ○ Orange

Your favourite color is Blue

5. Rewrite the above exercise to handle the events using DOM2 event model.

```
<!DOCTYPE html>
<html>

<body>

  <p>Select your favourite color:</p>

  <label><input type="radio" name="c" value="Red"> Red</label>
  <label><input type="radio" name="c" value="Blue"> Blue</label>
  <label><input type="radio" name="c" value="Green"> Green</label>
  <label><input type="radio" name="c" value="Yellow"> Yellow</label>
  <label><input type="radio" name="c" value="Orange"> Orange</label>

  <p id="out"></p>

  <script>
    const radios = document.getElementsByName("c");

    function show(color) {
      out.innerHTML = "Your favourite color is " + color;
    }

    radios.forEach(r =>
      r.addEventListener("click", () => show(r.value))
    );
```

```
  </script>

</body>

</html>
```

Output:

Select your favourite color:

○ Red ○ Blue ○ Green ○ Yellow ○ Orange

Select your favourite color:

○ Red ○ Blue ○ Green ◉ Yellow ○ Orange

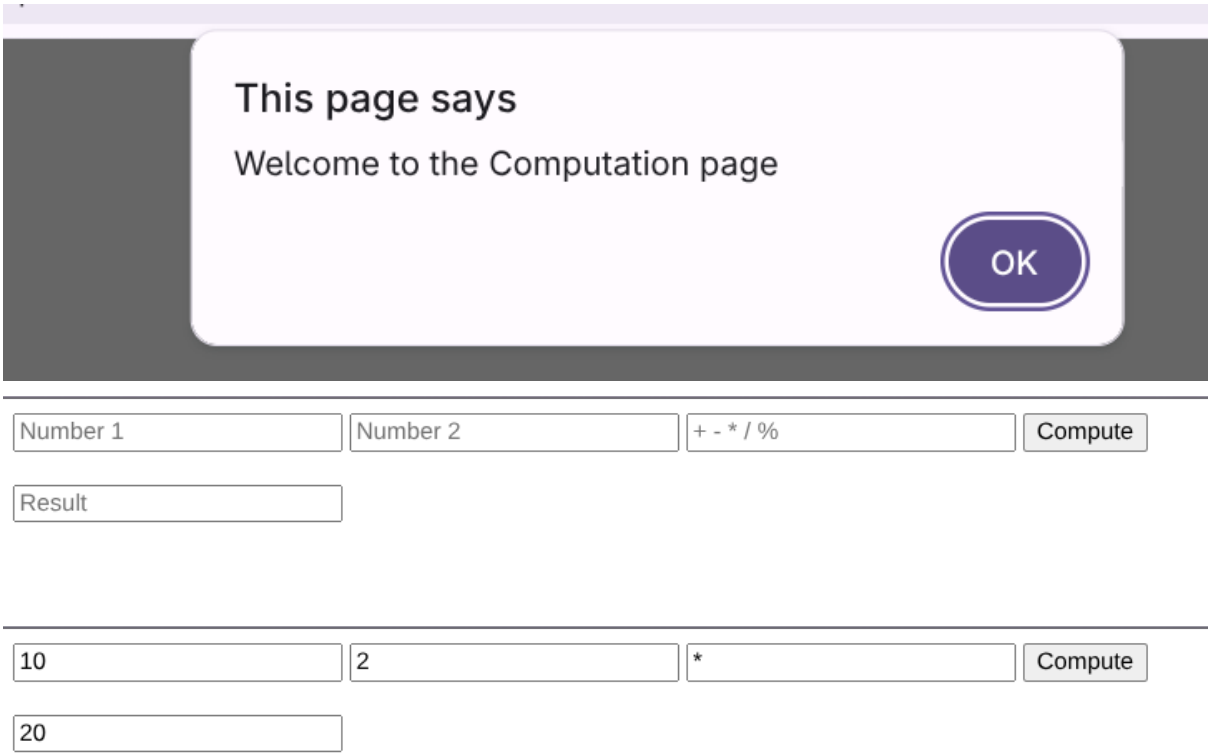
Your favourite color is Yellow

6. Create a JavaScript for the following: When a page is loaded, it should display 'Welcome to the Computation page' in an alert; then two numbers are provided in text boxes (Validate that they are numbers-if nonnumeric value is displayed, appropriate message should be displayed). In another text box, provide an operation to be carried out in the form of symbols (+,-,*,/,%). Validate this entry also. Include a button 'Compute' which when clicked, corresponding arithmetic operation should be carried out and result should be displayed in a text box(which should not be allowed to edit)

```
<!DOCTYPE html>
<html>
<body onload="alert('Welcome to the Computation page')">

<input id="n1" placeholder="Number 1">
<input id="n2" placeholder="Number 2">
<input id="op" placeholder="+ - * / %">
<button onclick="calc()">Compute</button><br><br>
```

```
<input id="res" placeholder="Result" readonly>
<p id="msg"></p>

<script>
function calc(){
 let a = n1.value, b = n2.value, o = op.value.trim();

 if(isNaN(a) || isNaN(b) || a=="" || b==="")
  return msg.innerHTML = "Enter valid numbers";

 if(![""+","-","*","/","%"].includes(o))
  return msg.innerHTML = "Invalid operator";

 a = +a; b = +b;
 let r = {"+":a+b,"-":a-b,"*":a*b,"/":a/b,"%":a%b}[o];

 res.value = r;
 msg.innerHTML = "";
}
</script>

</body>
</html>
```

Output:



10 | 2 | + | Compute

12

10 | 2 | - | Compute

8

10 | 2 | / | Compute

5

10 | 2 | % | Compute

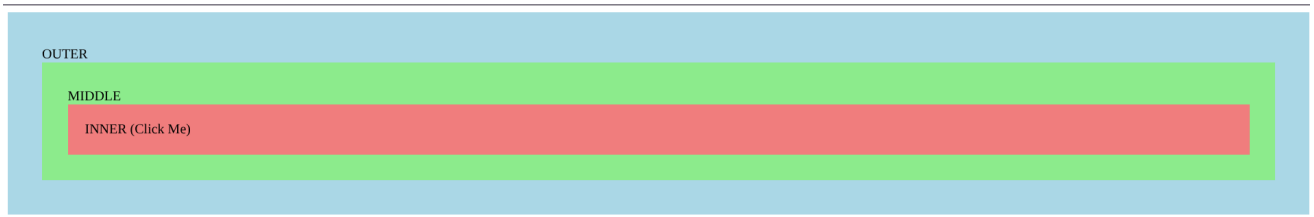
0

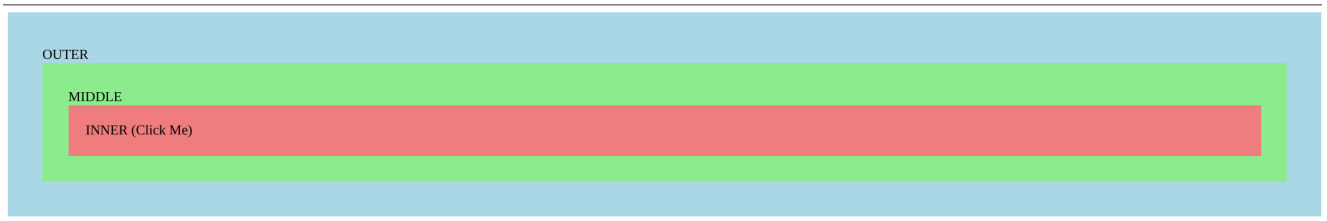
7. Show the concept of event propagation in JavaScript event handling.

```
<!DOCTYPE html>
<html>

<body>

  <div id="outer" style="padding:40px; background:lightblue;">
    OUTER
    <div id="middle" style="padding:30px; background:lightgreen;">
      MIDDLE
      <div id="inner" style="padding:20px; background:lightcoral;">
        INNER (Click Me)
      </div>
    </div>
  </div>

  <p id="log"></p>

  <script>
```

```
    function show(msg) {
      log.innerHTML += msg + "<br>";
    }

    // Bubbling (default)
    outer.addEventListener("click", () => show("Outer Bubble"));
    middle.addEventListener("click", () => show("Middle Bubble"));
    inner.addEventListener("click", () => show("Inner Bubble"));

    // Capturing
    outer.addEventListener("click", () => show("Outer Capture"), true);
    middle.addEventListener("click", () => show("Middle Capture"), true);
    inner.addEventListener("click", () => show("Inner Capture"), true);
  </script>

</body>

</html>
```

Output:





Outer Capture
Middle Capture
Inner Capture
Inner Bubble
Middle Bubble
Outer Bubble
Outer Capture
Middle Capture
Middle Bubble
Outer Bubble
Outer Capture
Outer Bubble
Outer Capture
Middle Capture
Inner Capture
Inner Bubble
Middle Bubble
Outer Bubble
Outer Capture
Middle Capture
Middle Bubble
Outer Bubble